

VLM Experiment: gpt-4o — Ordered vs. Shuffled Frame Classification

Model: gpt-4o | Frames per video: 4 | Videos evaluated: 25

Condition	Accuracy
Ordered Frames	68.00%
Shuffled Frames	44.00%
Drop (Ordered – Shuffled)	24.00pp

Case Studies — One Example per Action Class (Ordered Frames)

Case 1: Pushing [something] from left to right

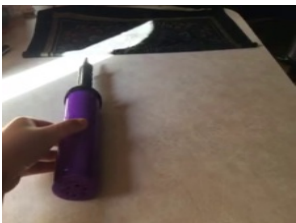
Video ID: 85446 | Ground Truth Label: 0 | Ordered Prediction: 1 (Wrong) | Shuffled Prediction: 3 (Wrong)



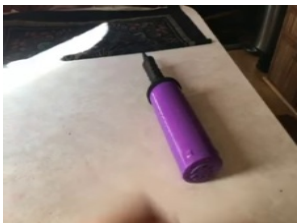
Frame 1



Frame 2



Frame 3



Frame 4

Model Chain-of-Thought (Ordered):

Step 1 - The motion observed across the frames shows a hand moving an object horizontally from the right side toward the left side.

Step 2 - Frame 3 provides the clearest evidence of direction as it shows the hand actively holding and moving the object.

Step 3 - Label 1 = a hand moves an object HORIZONTALLY from the RIGHT side toward the LEFT side

FINAL_LABEL: 1

Model Response (Shuffled):

Step 1 - The motion observed across the frames shows a hand moving an object vertically downward toward the surface.

Step 2 - Frame 4 provides the clearest evidence of direction as it shows the hand holding the object and moving it downward toward the surface.

Step 3 - Label 3 = a hand moves an object VERTICALLY DOWNWARD (toward the surface, toward the bottom of frame)

FINAL_LABEL: 3

Case 2: Pushing [something] from right to left

Video ID: 163169 | Ground Truth Label: 1 | Ordered Prediction: 1 (Correct) | Shuffled Prediction: 1 (Correct)



Frame 1



Frame 2



Frame 3



Frame 4

Model Chain-of-Thought (Ordered):

Step 1 - The motion observed across the frames shows a hand moving an object horizontally from the right side toward the left side.

Step 2 - Frame 3 provides the clearest evidence of direction as the object is noticeably further to the left compared to its position in Frame 1.

Step 3 - Label 1 = a hand moves an object HORIZONTALLY from the RIGHT side toward the LEFT side

FINAL_LABEL: 1

Model Response (Shuffled):

Step 1 - The motion observed across the frames shows a hand moving an object horizontally from the right side toward the left side.

Step 2 - Frame 4 provides the clearest evidence of direction as the object is positioned further to the left compared to its initial position in Frame 1.

Step 3 - Label 1 = a hand moves an object HORIZONTALLY from the RIGHT side toward the LEFT side

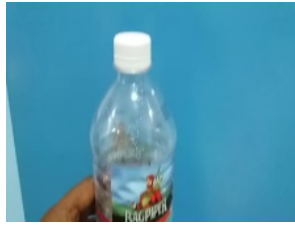
FINAL_LABEL: 1

Case 3: Moving [something] up

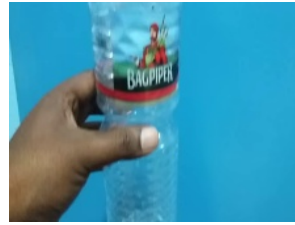
Video ID: 90063 | Ground Truth Label: 2 | Ordered Prediction: 3 (Wrong) | Shuffled Prediction: 3 (Wrong)



Frame 1



Frame 2



Frame 3



Frame 4

Model Chain-of-Thought (Ordered):

Step 1 - The motion observed across the frames shows a hand holding a bottle, which appears to be moving vertically downward.

Step 2 - Frame 4 provides the clearest evidence of direction as the bottle is shown at the bottom of the frame, indicating downward movement.

Step 3 - Select exactly one label from the numbered list below. Read each description carefully before deciding:

Label 3 = a hand moves an object VERTICALLY DOWNWARD (toward the surface, toward the bottom of frame)

FINAL_LABEL: 3

Model Response (Shuffled):

Step 1 - The motion observed across the frames shows a hand moving a bottle vertically downward.

Step 2 - Frame 4 provides the clearest evidence of direction as it shows the bottle at the bottom of the frame, indicating downward movement.

Step 3 - Label 3 = a hand moves an object VERTICALLY DOWNWARD (toward the surface, toward the bottom of frame)

FINAL_LABEL: 3

Case 4: Moving [something] down

Video ID: 155885 | Ground Truth Label: 3 | Ordered Prediction: 3 (Correct) | Shuffled Prediction: 3 (Correct)



Frame 1



Frame 2



Frame 3



Frame 4

Model Chain-of-Thought (Ordered):

Step 1 - The motion observed across the frames shows a hand moving an object vertically downward toward a surface.

Step 2 - Frame 4 provides the clearest evidence of direction as the object is closest to the surface, indicating a downward movement.

Step 3 - Label 3 = a hand moves an object VERTICALLY DOWNWARD (toward the surface, toward the bottom of frame)

FINAL_LABEL: 3

Model Response (Shuffled):

Step 1 - The motion observed across the frames shows a hand moving a ball vertically downward toward a surface.

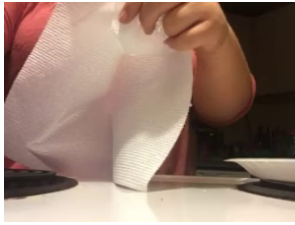
Step 2 - Frame 3 provides the clearest evidence of direction as the ball is closest to the surface, indicating downward movement.

Step 3 - Label 3 = a hand moves an object VERTICALLY DOWNWARD (toward the surface, toward the bottom of frame)

FINAL_LABEL: 3

Case 5: Tearing [something] into two pieces

Video ID: 7284 | Ground Truth Label: 4 | Ordered Prediction: 4 (Correct) | Shuffled Prediction: 4 (Correct)



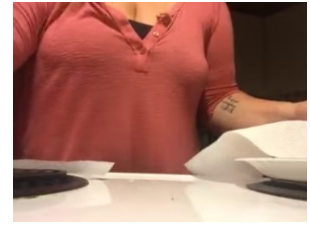
Frame 1



Frame 2



Frame 3



Frame 4

Model Chain-of-Thought (Ordered):

Step 1 - The motion observed across the frames shows a person tearing a paper towel into two separate pieces.

Step 2 - Frame 3 provides the clearest evidence of direction as it shows the paper towel split into two separate pieces.

Step 3 - Select exactly one label from the numbered list below. Read each description carefully before deciding:

Label 4 = a hand TEARS an object, splitting it into two separate pieces

FINAL_LABEL: 4

Model Response (Shuffled):

Step 1 - The motion observed across the frames shows a person tearing a paper towel, splitting it into two separate pieces.

Step 2 - Frame 2 provides the clearest evidence of direction as it shows the paper towel being separated into two pieces.

Step 3 - Select exactly one label from the numbered list below. Read each description carefully before deciding:

Label 4 = a hand TEARS an object, sp...